

Logistic Regression

- Supervised classification model.
- Dependent variable is categorical and binary (0 or 1)

Ind	Dep
Study Hour	Exam Result
0	0
2	0
3	0
4	0
5	1
6	1
7	1
8	1

Sigmoid $y = \frac{1}{1 + e^{-(a_0 + a_1 x)}}$

$a_0 = -1.5, a_1 = 0.6$
 $x = 5$

$y = 0.818$

